

Test Your Data Science Project with Google's Caliban

Data science projects need a real-time test environment to see if they are useful in real-life situations. Caliban solves this hurdle for a data scientist. It acts as a testing tool that tracks the environmental properties of your project during execution, and reproduces a specific running environment.



(Source: <https://github.com/google/caliban>)

Data science is an emerging field of advanced data analysis comprising collection, preparation, exploration, analysis, testing, visualisation, management and preservation of large data sets. In some cases, even a well-executed data science project may fail due to limitations in the test environment. Testing the project in all scenarios is mandatory to make it ready-to-use in reality. Caliban is a great open source tool developed by Google. It helps develop research workflows in isolated and reproducible learning environments. Caliban makes it easy to develop any machine learning prototype locally and run the code on

your device. A similar code can be tried in a cloud environment for execution on giant machines. It is best suited with JAX, TensorFlow and PyTorch for machine learning workflows.

Caliban requires Python 3.6 or above as a prerequisite. Docker is another important component. Install `nvidia-docker2` version, if you have a CPU and want to use Caliban.

Installing Caliban

Once the prerequisites are in place, Caliban can be installed using the following commands:

```
pipx install caliban
pip install caliban
```

The same commands can also be used to upgrade Caliban. After it's installed, you can check Caliban on the device by giving the following command:

```
caliban - help
```

This open source tool contains several useful sub-commands, which are essential for the execution of any project.

- **shell:** This generates a Docker image with the required dependencies. This environment is similar to what is available with the code.
- **notebook:** This starts a Jupyter notebook within a Docker image containing dependencies.
- **run:** This command packages directory code into a Docker image and executes it locally.
- **cloud:** This command allows submitting jobs to the cloud and runs inside the same Docker image that is used with the `run` command.
- **build:** This builds a Docker image used in the cloud and runs without submitting any code.
- **cluster:** This command creates Google Kubernetes Engine (GKE) clusters and submits jobs to them.
- **status:** This displays information about all the jobs submitted by Caliban. It is used to cancel pending jobs, re-build containers, or resubmit a batch after fixing errors.
- **stop:** This command stops running jobs.